

CAIRNWELL, United Kingdom

WMO: 030720

Lat: 56.8795N

Lon: 3.4213W

Elev: 933

StdP: 90.61

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-7.0	-13.8	1.3	1.1	-10.3	1.7	-3.0	32.0	-0.2	29.3	-1.0	13.7	350	1.208

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	16.5	12.1	14.8	11.4	13.5	10.9	13.7	14.9	12.6	13.7	11.7	12.7	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	10.5	14.1	12.1	9.8	13.0	11.1	9.2	12.0	41.3	15.0	38.3	13.7	35.8	12.9	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
36.4	28.8	24.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-9.4	14.7	1.3	1.1	-10.4	15.5	-11.1	16.2	-11.8	16.8	-12.8	17.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-1.5	-1.3	-0.6	1.6	4.9	7.0	9.2	8.5	7.0	3.9	0.4	-0.7		
	DBStd	4.86	2.71	3.21	3.26	3.39	3.50	3.02	2.94	2.46	2.54	2.74	2.78	3.00		
	HDD10.0	2531	356	316	329	252	165	99	52	61	97	188	287	330		
	HDD18.3	5510	614	549	587	502	417	340	283	306	341	446	537	588		
	CDD10.0	61	0	0	0	0	6	9	27	13	5	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	1	0	0	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0		
(14) Wind	WSAvg	11.1	12.5	12.7	12.1	10.5	8.6	7.7	7.6	9.4	10.9	12.7	13.2	15.0		
(15) Precipitation	PrecAvg	987	117	78	66	62	64	64	76	78	68	109	106	97		
	PrecMax	1268	236	199	144	151	118	128	152	185	176	189	197	278		
	PrecMin	641	12	20	11	21	17	20	22	11	16	21	39	27		
	PrecStd	135	55	45	25	33	30	28	34	42	41	46	45	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.0	10.2	12.3	16.6	17.6	18.8	17.9	15.1	12.2	9.2	8.6		
		MCWB	3.6	3.8	4.1	6.9	11.2	11.0	13.7	13.6	9.6	6.4	7.4	5.6		
	2%	DB	5.1	6.2	6.9	9.8	14.0	14.5	16.8	15.2	13.2	10.1	7.1	6.9		
		MCWB	4.6	4.5	3.7	6.6	10.2	10.4	12.7	12.8	11.2	9.5	6.6	5.1		
	5%	DB	4.0	4.6	5.1	8.0	12.0	12.8	15.1	13.4	12.0	9.0	5.8	5.4		
		MCWB	3.7	3.9	3.5	5.3	8.4	10.0	11.9	11.7	10.6	8.4	5.0	4.6		
	10%	DB	2.8	3.3	3.8	6.3	10.2	11.2	13.7	11.9	10.6	7.9	4.6	3.8		
		MCWB	2.5	2.8	2.9	4.5	7.0	9.1	11.4	10.8	9.3	7.5	4.0	3.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.1	5.9	9.0	12.8	13.1	14.9	15.5	13.4	10.4	8.4	7.7		
		MCDB	6.1	7.2	7.6	9.9	14.9	15.0	16.8	16.6	14.0	10.4	8.7	7.8		
	2%	WB	4.7	5.5	4.7	7.3	11.0	11.8	13.7	13.6	12.1	9.4	6.5	6.2		
		MCDB	4.8	5.7	5.8	8.7	13.3	13.0	15.2	14.4	12.5	9.6	6.8	6.4		
	5%	WB	3.6	4.2	3.8	5.9	9.1	10.6	12.7	12.1	10.7	8.3	5.2	4.8		
		MCDB	3.8	4.4	4.7	7.3	11.1	12.0	14.1	12.8	11.3	8.5	5.6	5.1		
	10%	WB	2.5	2.9	3.0	4.8	7.7	9.6	11.7	11.1	9.5	7.3	4.1	3.3		
		MCDB	2.8	3.1	3.8	6.0	9.5	10.8	13.1	11.8	10.1	7.5	4.5	3.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.4	3.5	3.7	4.3	5.3	4.9	4.7	4.1	3.9	3.5	3.4	3.7		
		MCDBR	5.3	5.1	5.1	6.4	6.5	6.7	6.0	5.3	5.2	5.1	5.5	5.8		
		MCWBR	5.0	4.3	3.9	4.5	4.3	4.7	4.1	4.1	4.7	4.9	5.3	5.6		
		MCDBR	5.2	4.6	5.0	5.4	6.4	5.7	5.2	4.6	4.9	4.6	5.5	5.6		
	5% WB	MCWBR	5.1	4.5	4.4	4.3	4.6	4.7	4.2	4.2	4.8	4.7	5.4	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.288	0.311	0.330	0.331	0.330	0.331	0.320	0.308	0.296	0.269	0.249		
	taud		2.299	2.421	2.405	2.402	2.435	2.452	2.446	2.485	2.532	2.520	2.383	2.227		
	Ebn at Noon		635	787	855	889	908	911	902	891	855	771	635	542		
	Edh at Noon		48	65	85	100	103	103	101	91	75	59	44	39		
(44) All-Sky Solar Radiation	RadAvg		0.47	1.11	2.14	3.51	4.71	4.64	4.49	3.55	2.49	1.33	0.61	0.34		
	RadStd		0.07	0.10	0.26	0.36	0.27	0.38	0.38	0.23	0.18	0.14	0.06	0.04		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (39 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page